

EE 208P: Digital Systems Design Practicum

Course Code	EE 208P
Course Name	Digital Systems Design Practicum
L-T-P-C	1-0-2-2
Prerequisites	Applied Electronics (IC161) or Equivalent
Intended for	UG
Distribution	Discipline Core for B. Tech EE (2nd year and above); Elective for B.Tecll in CE, CSE and ME (2nd year and above)
One-time approval	5th Senate (Oct 2013)
Replaced by	EE 210 by the 24th Senate (Feb 2020)

Course Contents

- **Introduction to digital systems:** Review of logic design, combinational and sequential digital system design
- **Introduction to Microcontrollers:** Introduction to microcontrollers, overview of architecture of a typical microcontroller such as AVR microcontroller, addressing, assembly language programming, interfacing with I/O devices, timer/counter programming, interrupt processing, etc.
- **Real world interfacing of microcontrollers:** Interfacing with simple devices such as LCD, keyboard, motor control, sensors, LED 7 segment display, DTMF decoder, etc.
- **Hardware Description Language:** Introduction to hardware description language, overview of structural, behavioral and dataflow modeling of digital systems using hardware description language, notion of finite state machines, delay modeling, memory modeling, synthesizable & non-synthesizable HDL codes for digital system design.
- **Introduction to FPGA:** Introduction to complex digital systems design, notion of programmable logic devices, overview of FPGA architectural realization of data path and controller, timing analysis of data-path and controller, synthesis, placement, routing, performance optimization,
- **FPGA based systems design:** Implementation of simple systems using FPGA exercising the timing closure paths.
- **Physical design automation:** Partitioning, floor-planning, placement, routing; clock design considerations, timing margins, clock skew, clock distribution networks.

Course lectures

The course lectures will comprise of 1 hour of lecture per week covering the fundamentals of the topics.

Mini project

The students will be required to carry out a small project on topics such as but not limited to Electronic logic, Voting machine, Traffic Light controller, Automatic Room Light Controller, etc. The projects are aimed at exposing the students to real life design issues.

Text Books

1. Dhananjay Gadre, **Programming and Customizing the AVR microcontroller**, Tata McGraw Hill, 2014.
2. Wayne Wolf, **FPGA based Systems Design**, Pearson Education, 2003.
3. Stephen Brown and Zvonko Vranesic, **Fundamentals of Digital with VHDL Design**, McGraw Hill, 2008.
4. Volnei A. Pedroni, **Circuit Design with VHDL**, The MIT Press, 2004.

References

1. Steve Kilts, **Advanced FPGA Design: Architecture, Implementation and Optimization**, J. Wiley and Sons, 2007.
2. Seetharaman Ramachandran, **Digital VLSI Systems Design**, Springer Verlag, 2012.
3. Peter Ashenden, **The designer's guide to VHDL**, Morgan Kaufmann, 2008.
4. Charles H. Roth Jr., **Digital Systems Design using VHDL**, Cengage Learning, 2014.